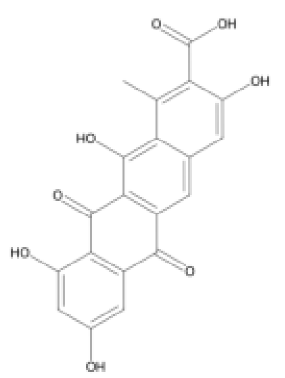
Figure 4. Qualitative showcase across vision, homology, and multi-task generation

A struct_cap — list functional groups



Prompt

Please list the functional groups of the molecule.

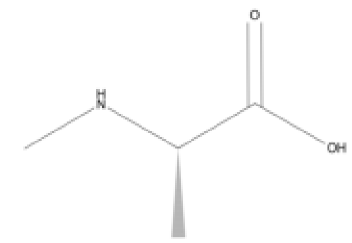
Reference

This molecule contains several functional groups, including ketones, alcohol groups, carboxylic acids, side-chain hydroxyls, side-chain aldehydes, benzene, and naphthalene.

v3 prediction

The molecule shown contains the following functional groups:
* Carboxylic acid: Represented by the -COOH group at the top of the structure.
* Phenol: Multiple hydroxyl groups (-OH) are attached directly to aromatic rings, which is characteristic of phenols.
* Ketone: Two carbonyl groups (C=O) are part of the ring system, specifically within the anthraquinone-like core. ...

B general_desp — describe the molecule



Prompt

Please describe this drug.

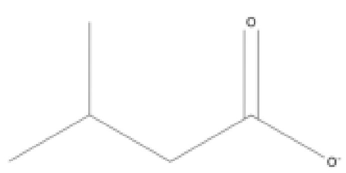
Reference

The molecule is a methyl-L-alanine in which one of the the amino hydrogen of L-alanine is replaced by a methyl group. It is a tautomer of a N-methyl-L-alanine zwitterion.

v3 prediction

This image shows the chemical structure of (S)-2-amino-3-methylbutanoic acid, also known as L-valine. It is a nonessential amino acid that is important for the biosynthesis of proteins. It is found in many foods, including eggs, cheese, and meat. It is also used in the production of various pharmaceuticals and other chemical products.
...

C trans_iupac — IUPAC naming



Prompt

What is the IUPAC of the molecule?

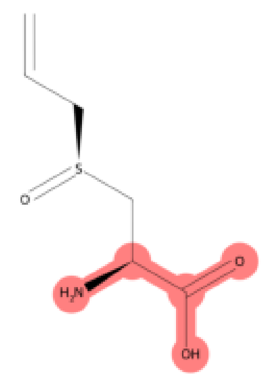
Reference

The IUPAC is 3-methylbutanoate.

v3 prediction

The IUPAC name for the molecule shown is 3-methylbutanoate.

D struct_recog — highlighted group



Prompt

Which functional group is highlighted?

Reference

The functional group amino acids is highlighted.

v3 prediction

The highlighted functional group is a carboxylic acid.

In the provided image, the carboxylic acid group is part of the amino acid structure, specifically within the context of the molecule's overall structure. The highlighting clearly identifies the -COOH group, which is characteristic of carboxylic acids.

E standard homology — positive pair

Sequence 1

`MTMIKICGLSTPETIEAAVQAGATHVGLVHFAKSPRHVELEKAAELRALVPESVKAVLLLVNEQPEETARAIQIVKPDVV...`

Sequence 2

`VKICGLKTAEAVEHAVSLGASHTGFIFFPKSPRNIEPDDAGRLAERIRGRAKIVAVTVDADNDDLDEIVSALKPDILQLH...`

Gold label

Homologous

v3 prediction

Homologous

✓ correct

F remote homology — positive pair

Sequence 1

`atklteekyelkegqtldvkcdytlekfassqkawqiirdgempktlacterpsknshpvqvgriiledyhdhgllrvrm...`

Sequence 2

`qvqlvqsggqmkkpgesmriscrasgyefidctlnwirlapgkrpewmgwlkprggavnyarplqgrvtmtrqlsqdpdd...`

Gold label

Homologous

v3 prediction

Homologous

✓ correct

G cell type from markers

Instruction

Infer the most likely cell identity from these markers.

Input

`IAPP, NKX6-1, MAFA, INS`

Reference

This cell is most consistent with pancreatic beta cell, a pancreatic beta cell secreting insulin. Key markers such as INS, IAPP, NKX6-1 support this identity.

v3 prediction

This cell is most consistent with pancreatic beta cell, a pancreatic beta cell secreting insulin. Key markers such as NKX6-1, INS, IAPP support this identity.

H physico-chemical descriptor from SMILES

Instruction

Describe key molecular descriptors for the given SMILES.

Input

`CNC(=O)c1ccc(C)c(-n2c(C)cc(OCc3ccc(F)cc3F)c(Br)c2=O)c1`

Reference

This molecule is medium-sized and moderately polar, with lipophilic character. It contains aromatic ring, amide, ether, halide, pyridine. Its descriptor profile: MW=477.31, LogP=4.43, TPSA=60.33, HBD=1, HBA=3, QED=0.59.

v3 prediction

This molecule is medium-sized and moderately polar, with lipophilic character. It contains aromatic ring, halide, amide, ether, pyridine. Its descriptor profile: MW=532.39, LogP=5.38, TPSA=75.4, HBD=1, HBA=5, QED=0.49.